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Research progress of mRNA vaccines for infectious diseases

Zhen Zhang^{1,2†}, Jingli Du^{1†}, Danyang Zhang^{1†}, Rui Han^{2*}, Xueqiong Wu^{1*} and Yan Liang^{1*}

Abstract

mRNA vaccines for infectious diseases prevent diseases by stimulating the body to produce specific immune responses through mRNA molecules encoding pathogenic proteins. Compared to traditional vaccines (e.g., inactivated, live attenuated, subunit, recombinant protein and viral vectors), mRNA vaccines offer several advantages including high safety, potent efficacy, scalable large-scale production, and cost-effectiveness. mRNA vaccines have demonstrated significant potential in combating infectious diseases since their inception. In particular, during the 2019 Coronavirus Disease (COVID-19) pandemic, the mRNA vaccines delivered with lipid nanoparticles (LNPs) have been developed by BioNTech and Moderna, their exceptional protective efficacy and extensive clinical application further proved the rapid responsiveness of mRNA vaccines in addressing emerging infectious diseases. This success has brought mRNA vaccines back into the spotlight of the scientific community. This article reviews the molecular biological basis, delivery systems, and immune mechanisms of mRNA vaccines, as well as the progress of research and clinical trials related to mRNA vaccines targeting the COVID-19 virus, influenza virus, rabies virus, Zika virus, human immunodeficiency virus, and *Mycobacterium tuberculosis* (*M.tb*), while also discussing the current challenges faced in the application of mRNA vaccines. These discussions provide a theoretical foundation and practical guidance for the future development of mRNA vaccines targeting bacterial infectious diseases such as tuberculosis (TB).

Keywords mRNA vaccines, MRNA vaccine components, Delivery systems, Immune mechanisms, Clinical trials

Introduction

Vaccines have played a critical role in global public health as a powerful weapon against infectious diseases caused by bacteria or viruses. They have evolved through three generations: the first generation of traditional vaccines (including attenuated live vaccines, inactivated vaccines,

and toxoid vaccines), the second generation of genetically engineered vaccines (mainly subunit vaccines, synthetic peptide vaccines, and recombinant genetically engineered live vaccines), and the third generation of vaccines (mainly including DNA vaccines, recombinant viral vector vaccines, and mRNA vaccines) [1]. Vaccine research has increasingly demonstrated its important role in disease prevention and treatment [1].

Traditional vaccines have demonstrated a notable preventive effect against infectious diseases, such as smallpox and poliomyelitis. However, it is necessary to take into consideration their associated risks, and research, development, and production cycles, as well as the limitations in aspects, such as scalability and adaptability, to new pathogens [2]. Second-generation recombinant vaccines have also played a crucial role in reducing epidemics such as hepatitis B. However, they usually require a suitable expression system and the combined use of

[†]Zhen Zhang, Jingli Du and Danyang Zhang contributed equally to this study.

*Correspondence:

Rui Han

shhr950822@163.com

Xueqiong Wu

xueqiongwu@139.com

Yan Liang

amy5919@sina.com

¹ Beijing Key Laboratory of New Techniques of Tuberculosis Diagnosis and Treatment, Institute of Tuberculosis Research, Senior Department of Tuberculosis, The Eighth Medical Center of PLA General Hospital, Beijing 100091, China

² Graduate School, Hebei North University, Zhangjiakou, China



effective adjuvants to be effective [3]. The mRNA vaccine in the third generation of vaccines has played a crucial role in ending the COVID-19 pandemic. Compared to traditional vaccines (e.g., inactivated, live attenuated, subunit, recombinant protein, and viral vectors), mRNA vaccines offer many advantages, including rapid design and synthesis, the ability to adapt the sequence based on viral mutations or the structure of emerging pathogens, and rapid large-scale production with high cost-effectiveness. Importantly, mRNA vaccines are non-infectious and do not integrate into the host's genome, effectively avoiding the risk of potential mutations and ensuring safety.

In 1961, mRNA was identified as an intermediary that carries genetic information to ribosomes for protein synthesis. With the development of in vitro transcription (IVT) technology for RNA production, synthetic mRNAs can be engineered to resemble mature and processed mRNA molecules and scientists have begun to use mRNAs to express desired proteins [4]. Initially, the use of mRNA to express proteins in vivo was limited by its instability and the genetically inflammatory nature of unmodified nucleotides. However, with the rapid advances in delivery technology and the development of nucleoside modification, sequence/codon optimization, and design of the UTR techniques to address these issues, mRNA technology has rapidly expanded in biomedical applications [5]. The major milestone in the development of mRNA vaccines was the widespread clinical use of two rapidly developed and licensed COVID-19 mRNA vaccines by BioNtech/Pfizer and Moderna during the COVID-19 pandemic in 2021, which showed significant preventive effects. The mRNA vaccines using lipid nanoparticles (LNPs) as delivery materials overcome the inherent limitations of bare mRNA molecules [6].

Although mRNA vaccines are still being administered, adverse reactions to them are constantly being reported [7] and they still have the following limitations: (1) Storage and transportation requirements: The strict ultra-low-temperature cold-chain requirements increase costs and affect accessibility, posing challenges to regions with weak infrastructure [8]. (2) Short-term adverse reactions: Including local pain and systemic symptoms, such as fatigue and fever. The probability of severe allergic reactions is extremely low, but emergency treatment is required [9]. (3) Long-term safety: The application of new nanotechnology lacks long-term follow-up data. Although there is no evidence showing an impact on fertility or the development of chronic diseases [10]. (4) Immune tolerance differences: There is heterogeneity in the response of individuals to mRNA vaccines. For instance, the antibody production in the elderly or those with weakened immunity is relatively weak. Therefore,

adjuvants (such as TLR agonists) or multivalent vaccines need to be developed to enhance the response [11]. (5) Scalability: The production process of mRNA vaccines involves multiple steps, including mRNA synthesis, purification, and formulation with delivery vehicles (such as lipid nanoparticles). Each step requires strict quality control, and any disruption in the supply chain of raw materials or equipment can affect production capacity.

Currently, research on mRNA vaccines for various diseases has grown exponentially, and mRNA vaccines for some infectious diseases have entered clinical trials. mRNA vaccine has brought new hope for controlling these diseases [5]. Tuberculosis (TB) is currently an infectious disease with high morbidity and mortality, and the Bacillus Calmette–Guérin (BCG) vaccine remains the only licensed vaccine for human use in the prevention of tuberculosis. Newborns vaccinated with BCG can reduce their risk of *Mycobacterium tuberculosis* (*M.tb*) infection up to the age of 10 years [12]. However, the protective efficacy of BCG vaccination in adults varies widely, ranging from 0 to 80% [13]. Therefore, there is an urgent need to develop a TB vaccine with high protective efficacy. The emerging mRNA vaccine technology undoubtedly offers new hope for TB vaccine research. Therefore, this paper will review the current progress of mRNA vaccine research in terms of structure, type, immunogenicity regulation, delivery methods, mechanism of inducing immune response, and mRNA vaccines for various diseases that have entered into clinical trials, as well as put forward the current challenges and look forward to the possible future research directions, providing a theoretical basis for future mRNA vaccine research.

mRNA vaccine structural elements

mRNA structure and its optimization

Currently, mRNA IVT technology has reached a relatively advanced stage, with the most common methods using phage T3, T7, or SP6 RNA polymerase and linearized DNA for mRNA synthesis [14]. The conventional mRNA has a specific structure, including: 5'-end cap structure (5' cap), 5'-end untranslated region (UTR), protein-encoding open reading frame (ORF), 3'-end translated region, and the polyadenylated poly(A) tail [15]. To promote the structural integrity and translational capability of mRNA vaccines, it is important to consider proper sequence design as an effective strategy to improve their stability. This involves enhancing the 5' end (e.g., through chemical modification of the 5' cap and incorporation of a shorter 5' UTR), refining the ORF region (e.g., by codon optimization and preventing the formation of excessively strong stem-loop structures within the ORF), and improving the 3' end (e.g., by introducing an AU-rich sequence into the 3' UTR, selecting

the optimal poly(A) tail length), and implementing nucleotide modifications.

5' cap

The chemical modification at the 5'-end of the mRNA molecule, known as the 5'-cap structure, which is formed gradually during the transcription and processing of mRNA is not derived directly from the coding sequence of the DNA template. The core 7-methylguanosine (m7G) nucleotide represents a non-template-dependent addition linked to the first nucleotide encoded by the template via a distinctive 5'-5' triphosphate bond. It plays a multitude of critical roles in molecular biology. For instance, during cytoplasmic mRNA translation process, it facilitates the recognition and binding of ribosome to mRNA, thereby participating in the initiation of protein translation and enabling the onset of protein synthesis. Additionally, the 5' cap collaborates with the 3'-end poly(A) tail, poly(A)-binding proteins, RNA helicase, and other initiation factors (such as eukaryotic initiation factor 4E (eIF4E)) to enhance translation efficiency [5, 16]. At the same time, the 5' cap structure protects the mRNA from degradation by the 5'-3' exonuclease. In addition, the 5' cap converts 5'-triphosphate (5'-PPP) into m⁷GpppN through capping enzymes, such as Vaccinia Capping Enzyme, or co-transcriptional capping enzymes, such as CleanCap. This process involves the coverage of free phosphate groups, thereby preventing RIG-I-mediated immune activation [19], reducing immunogenicity, and enhancing RNA stability.

The role of the mRNA 5' cap has contributed to the development of mRNA vaccines and their use in disease prevention and treatment. There are two main methods of capping during mRNA transcription *in vitro*: enzymatic capping (post-transcriptional) and co-transcriptional capping, but they do not occur simultaneously during IVT. Enzymatic capping uses RNA 5'-triphosphatase (RTPase) and guanylyl transferase (GTase) to form a 5'-cap structure. RTPase hydrolyzes the 5' γ -phosphate of mRNA to form β -phosphate, and then β -phosphate transfers guanosine mono-phosphorus (GMP) to the 5'-diphosphate RNA under the action of GTase to form the GpppNp-RNA structure. Guanine-N7 methyltransferase (N7MTase) methylates the guanosine moiety of the cap to form the Cap0 structure (m7GpppNp). The Cap0 structure is essential for the efficient translation of mRNAs as it serves as the basic form which protein factors bind and drive related functions. However, the Cap0 structure mRNA is perceived by the host as exogenous RNA, thereby triggering an inflammatory response by the host's natural immune system [17]. The formation of the Cap1 structure (m7GpppN1mp) by 2'-O-methylation of the first nucleotide in the mRNA in the presence of

2'-O-methyltransferase (2'-O-MTase) effectively reduces the immunogenicity of mRNAs in cells [18]. In addition, it assists cellular sensors and the type I interferon (IFN) signaling pathways in distinguishing the Cap1 structure from other exogenous RNAs, thereby preventing activation of the innate immune response [19]. Moderna's COVID-19 vaccine, known as mRNA-1273, uses this approach to cap the mRNA [20]. Further methylation at the 2'-hydroxyl position of the second nucleotide at the 5'-end of the mRNA results in the formation of the Cap2 structure (m7GpppN1mpN2mp) [21]. Recent studies have shown that while the Cap1 structure does indeed significantly reduce the immunogenicity of mRNAs, cellular mRNAs with only Cap1 type can still activate inflammatory antiviral mechanisms, while Cap2 can provide crucial additional protection [22]. Although Cap 2 is present in 10–15% of eukaryotic cells, its role is not fully understood. Enzymatic capping is characterized by transcription followed by capping, with a capping rate of almost 100% [26]. However, the cost of various enzymes is high, the production process involves many steps with complicated quality control, and the enzymatic (post-transcriptional) capping is also time-intensive.

The second method is co-transcriptional capping, which uses capping analogs (such as the first generation of mCap, the second generation of ARCA, and the third generation of Cap1 cap structural analogs) to achieve co-transcriptional capping of mRNA [21, 23]. Currently, the CleanCap[®] capping analog products developed by Tri-Link Company can co-transcribe with the target mRNA to form the Cap1 structure with a capping rate of more than 90%, effectively solving the problems of low efficiency (a capping rate of co-transcriptional capping is less than 70%) and high enzyme cost (Enzymatic capping) in the traditional capping methods [24]. BioNTech's COVID-19 mRNA vaccines (BNT162b1 and BNT162b2) used CleanCap[®] capping analogs for vaccine capping [25]. Co-transcriptional capping is currently the most widely used method in IVT. It is characterized by one-step cap addition, simple operation and high success rate of capping, which can rapidly increase the yield of mRNA vaccines. However, the current cost of synthesizing cap analog is relatively high. Newly introduced Cap2 cap structural analogs, such as 3'-O-Me-m7G(5')ppp(5') (2'OMeG)p(2'OMEG), can be synthesized and capped more efficiently, with lower immunogenicity and higher protein translation efficiency [26].

UTR

UTRs (5'UTR and 3'UTR) are non-coding regions at the 5' and 3' ends of mRNA molecules. They can affect mRNA translation, half-life, and mRNA localization in cells. The 5'UTR is mainly derived from genes, such as

globin, heat shock protein 70 (Hsp70), and hydroxysteroid dehydrogenase (3 β -HSD), prevalently [27, 28]. Its main role is to engage in the translation of downstream ORFs [29, 30]. The 3' UTR is mainly derived from genes encoding the hemoglobin subunits α (HBA) and β (HBB), as well as the genes for albumin (Alb), heat shock protein 70 (Hsp70), tyrosine hydroxylase (TH), and collagen α 1(COL1A1) [31, 32]. Their main function is to maintain mRNA stability and improve translational efficiency [33, 34]. UTRs are optimized to enhance mRNA stability and improve the translation efficiency. Therefore, when designing mRNA, it is crucial to avoid placing the upstream open reading frames and start codons (AUG) within the 5'UTR, which prevents expression repression and mRNA degradation. In addition, the 5'UTR should not be too long so as not to a secondary structure that makes it difficult to bind to the ribosome [35]. Meanwhile, the Kozak sequence was added to the 5'UTR region to enhance the start codon recognition ability and improve the translation efficiency [36]. The AU-rich elements (AREs) in the 3'UTR are *cis*-acting regions that interact with ARE-binding proteins to regulate mRNA stability and steady-state expression [44]. The number and arrangement of AUUUA repetitions or UUAUUUAUU (U-rich) or AU-rich (AUAUAU A) in the AREs have a big effect on how the poly(A) tail is cut off and how the RNA breaks down afterward. GU-rich sequences (GREs) interact with the CELF1 protein in human cells, facilitating mRNA degradation. Consequently, the proper modification of AREs and GREs can preserve mRNA stability [37–39].

ORF

The ORF is a crucial segment in mRNA that encodes proteins and influences mRNA stability and translational efficiency. Optimizing the ORF primarily involves substituting uncommon codons in the foreign mRNA with the prevalent codons of the host cell, thereby reducing the presence of rare codons and enhancing translation efficiency. Certain research indicates that CureVac (a biopharmaceutical company headquartered in Germany) substitutes A or U in the third nucleotide of the CVn-CoV codon with C or G to enhance the GC content of the ORF, thereby improving the stability and translation efficiency of the mRNA [40]. Not all proteins necessitate outstanding translation efficiency, some require reduced efficiency to achieve proper folding [41]. Consequently, selecting a suitable optimization approach to equilibrate the translation rate and precision is essential.

Poly(A) tail

The poly(A) tail is an extended sequence of poly-adenosine. The length of the poly(A) tail influences mRNA

stability and translation efficiency [42]. It inhibits mRNA degradation and establishes a closed loop with eIF4E and the 5' cap to augment mRNA stability [43]. However, this tail is unstable and can be lost or cut off. Currently, there are two methods to polyadenylate mRNA: the first is the traditional enzymatic polyadenylation, which is unable to change the length of the tail [16], and the second is to design specific poly(A) sequences on DNA templates, which are then transcribed into poly(A) tail whose length can be changed [44]. A truncated poly(A) tail is incapable of promoting mRNA translation. Consequently, the ideal length of the poly(A) tail sequence enhances the translation efficiency and stability of mRNA [45, 46]. In a recent study, the poly(A) tail sequences were separated and connected by spacer sequences in the middle which could significantly reduce the recombination events during plasmid DNA amplification, maintain the tail length, and at the same time did not affect the translation efficiency and half-life of the mRNA transcribed in vitro [47]. Using branched poly(A) tails and the multiorgan protection sequence, these two key technological innovations in the development of linearly modified mRNA (modRNA) vaccines have significantly enhanced the stability, expression efficiency, and tissue targeting capabilities of traditional mRNA vaccines [55, 56].

Types of mRNA vaccines

Non-replicating mRNA vaccines

Non-replicating mRNA vaccines only encode the target antigen and do not contain replicase genes. Their diminutive size (about 2–3 kb), uncomplicated structure, and high specificity of the immune response make them distinctive. Currently, researchers have established two categories of non-replicating mRNA vaccines: the unmodified (natural) mRNA vaccination and the modified mRNA vaccine (Fig. 1A and B). The last one uses 1-methylpseudouridine or pseudouridine instead of uridine nucleoside, which might make mRNA work better at translation and make cells more stable [48]. Compared to self-amplifying mRNA, constrained expression of non-replicating mRNA requires higher doses to achieve the desired vaccination effect, or a comprehensive optimization process to elicit an effective immune response at reduced doses [49, 50].

Self-amplifying RNA (saRNA) vaccines

Non-replicating mRNA vaccines are extremely simple to transfect in vitro, nonetheless, they necessitate high dosages or numerous vaccinations to provoke a sufficient immune response. As an alternative technology addressing these concerns, saRNA vaccines mimic alphaviruses by delivering genetic information encoding the target antigen and other genes (e.g., viral RNA-dependent RNA

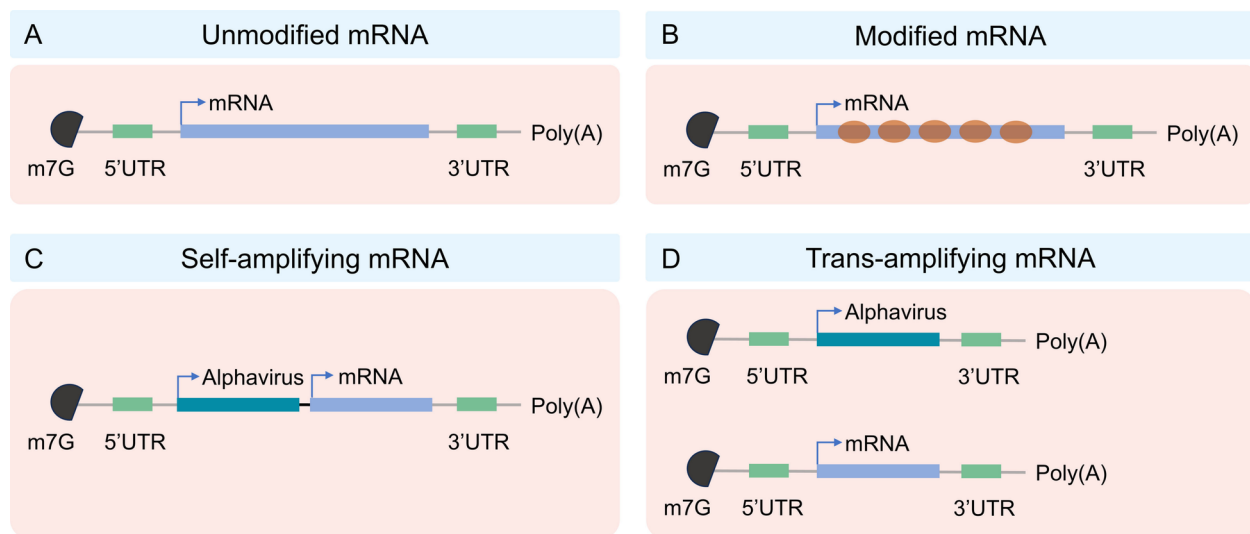


Fig. 1 Types of mRNA vaccines

polymerase), and are self-amplifying, autonomously replicating within host cells to significantly enhance target protein expression [51] (Fig. 1C). This technology is expected to trigger a strong immune response at low doses and prolong protection [52–55]. Moreover, saRNA can diminish the reliance on delivery materials (e.g., LNP), hence reducing expenses (Less LNP, lower dose) and adverse consequences (e.g., LNP toxicity, unmodified mRNA may be recognized by RIG-1/MDA5 to elicit an interferon response) [65]. These benefits could facilitate the vaccination of a greater number of individuals in a reduced timeframe during a pandemic [56]. In 2021, British scientists executed the inaugural clinical study of the novel saRNA COVID-19 vaccine (COVAC1). The findings indicated that 87% of vaccinated individuals elicited a protective immunological response at a modest dosage, with no safety concerns reported [57]. Recently, CSL and Arcturus Therapeutics collaborated to produce the novel coronavirus vaccine ARCT-154, the first commercially authorized saRNA vaccine for adult COVID-19, suitable for both primary and booster vaccinations in adults [58]. The saRNA vaccine has emerged as a leader in the next generation of RNA vaccines.

However, saRNA/replicons also have their limitations. The body may generate T cell responses against the replicase, and repeated injections may lead to the suppression of its expression. This severely limits the potential of saRNA vaccines for multiple booster immunizations and is one of the main obstacles in their clinical application. Furthermore, since the replicase cannot recognize many modified bases, replicons have long been limited to using natural uridine, which is a compatibility issue for base modification [59]. However, two recent reports showed

that saRNA can be modified with other nucleotides, such as 5-hydroxymethylcytidine, 5-methylcytidine (5-mC), and 5-methyluridine [60, 61]. In comparison to conventional mRNA, saRNA possesses a more extensive molecular structure and exhibits a reduced yield during in vitro synthesis. To overcome this technical limitation, Beisert et al. created an enhanced saRNA, known as trans-amplified RNA (taRNA), in 2020 (Fig. 1D). This is a novel *trans*-amplified RNA system characterized by excellent translation efficiency and minimal cellular translation interference, representing a significant advancement in RNA technology. This method makes the replicase complex work better by working together with two vectors: one that contains the viral immunogen and the other that contains the alphavirus replicon. Experimental findings confirm the use of taRNA in influenza hemagglutinin antigen-encoding RNA, showing that a minimal dose of 50 ng can significantly enhance the expression of the viral immunogen, thereby inducing an immunological protective response [62]. It is noteworthy that taRNA can solely encode the replicase, bypassing the length limitation of the target gene, thereby diminishing the risk of recombinant viral particles and enhancing safety [63]. This taRNA vaccination technology is adept at addressing rapidly evolving pathogens and new infectious illnesses [62].

Circular RNA (circRNA)

circRNAs are generated through back-splicing, wherein their 3' and 5' ends become covalently linked, forming a closed circular structure devoid of free termini. This unique topology constitutes the fundamental basis for circRNAs stability. Cellular exonucleases primarily recognize and cleave the free ends (5' or 3') of linear

RNAs [64, 65]. Owing to the absence of free ends, circRNAs exhibit significant resistance to degradation by these enzymes. Experimental evidence confirms that circRNAs possess marked resistance to RNase R, a potent 3'→5' exoribonuclease that efficiently degrades linear RNAs [66, 67].

This covalently closed circular architecture confers exceptional stability, enabling sustained and prolonged expression of antigenic proteins in vivo and offering a transformative solution to the transient expression profile inherent to linear mRNA therapeutics. Consequently, circRNA technology lays a solid foundation for developing next-generation RNA vaccines and therapeutics with enhanced efficacy (potentiating stronger immune responses), improved dosing convenience (potentially requiring lower doses and fewer administrations), and broader applicability (e.g., in oncology and therapeutic protein delivery).

Delivery systems for mRNA vaccines

Nucleases readily destroy mRNA, which is too large and negatively charged to pass through cell membranes. Research indicates that the absorption rate of naked mRNA molecules by cells is below 1 in 100,000 [5]. Administering mRNA vaccines using designated vectors is necessary to achieve optimal efficacy. The vector of choice must successfully penetrate the target cell membrane, internalize within the cell, avoid the endosome, release the mRNA into the cytoplasm, and aid in the translation process. At the same time, mRNA vaccine delivery vectors can change both the amount and quality of antigen protein production, which in turn changes how the host immune system responds [68]. This discussion will address several delivery systems, including lipid vector derivatives, polymers and protamine.

Lipid carrier derivatives

Lipid nanoparticles (LNPs)

The primary purpose of LNP, a lipid-based nanoparticle, is to deliver mRNA to the cytoplasm. By joining with the lipid bilayer of the early endosome, these nanoparticle carriers can move mRNA into the cell quickly and efficiently. This procedure not only liberates the mRNA into the cytoplasm but also safeguards it against degradation by ribonucleases (RNases) in systemic circulation [69]. LNP is mostly made up of four main parts: ionizable lipids (40–50%), cholesterol (38–45%), auxiliary phospholipids (10–12%), and polyethylene glycol (PEG) lipids (1–2%) [70]. These components collaborate to encapsulate and safeguard unprotected mRNA [71]. Ionizable amino lipids facilitate the release of mRNA from the endosome by interacting with the endosomal membrane [72]. PEG can make lipid nanoparticles stay in circulation

longer, stop mRNA from joining with plasma proteins, and speed up the clearance process. Phospholipids and cholesterol contribute to the stabilization of lipid nanoparticle structure [73].

But endosomal escape, which is necessary for RNA expression, also directly triggers inflammation by causing endosomal membrane damage. However, inflammation can be controlled by inhibiting galectins (large hole detectors) or using biodegradable lipids, which create smaller holes that are repairable by the endosomal sorting complex required for transport (ESCRT) pathway [74]. In addition, the amine headgroups in ionizable lipids drive LNP immunogenicity by binding to Toll-like receptor 4 and CD1d and by promoting lipid-raft formation. Immunogenic LNPs favor a type-1 T helper cell-biased immune response marked by increase in the immunoglobulins IgG2c and IgG1 and in the pro-inflammatory cytokines tumor necrosis factor, interferon γ , and interleukins IL-6 and IL-2 [75].

The delivery efficiency of LNP primarily relies on cationic or ionizable lipids, such as DLin-DMA, DLin-KC2-DMA [76], and DLin-MC3-DMA [77], which have been meticulously designed and synthesized. Additionally, C12-200 [78] and cKK-E12 [79] have been identified through high-throughput screening in combinatorial libraries. The next generation of ionizable lipids includes derivatives, such as DLin-MC3-DMA L319 [80], C12-200 and cKK-E12 [81–83], ALC-0315, and SM-102, which are used in COVID-19 vaccines [84], TT3 and its biodegradable derivative FTT5 [84, 85], vitamin-derived lipids ssPalmE [86] and VcLNP [87], A9 [88], L5 [89], A18 Lipid [90], ATX Lipid [91] and LP01 [92]. Most of these lipids have the capability to be biodegraded.

DLin-MC3-DMA has been approved for clinical treatment of RNA by ionizing cationic lipid [93]. Moderna used DLin-MC3-DMA ionizable lipids to make mRNA vaccines against the Zika virus and influenza. They then conducted both preclinical and clinical trials on these vaccines [94, 95]. Subsequent studies revealed the breakdown of the dialkyl tail of the dilauryl lipid in the ionizable lipid, suggesting that repeated administration of the drug could potentially increase its toxicity over time [96]. Consequently, Moderna has created a novel ionizable lipid, Lipid H, SM-102. The novel lipid features an enlarged branching structure at the tail end, and the incorporation of an ester bond improves biodegradability [89, 97]. Additionally, BioNTech has used the lipid nanoparticle form of its COVID-19 mRNA vaccine along with the ionizable lipid ALC-0315, which has a chemical structure similar to SM-102 and has also shown good results [96]. G-LNPs, which are lipid-modified poly-(guanine-lipoic acid) polymers, are a new type of lipid nanoparticle for mRNA vaccines that has been discovered recently.

G-LNPs can greatly improve the translation efficiency of mRNA that is encapsulated and reduce inflammation after vaccination by blocking reactive oxygen species that slow down translation and cause inflammatory reactions. Consequently, this markedly enhances the translation efficiency of the administered mRNA and diminishes inflammation post-vaccination. In vivo studies show that G-LNPs deliver mRNA effectively, triggering a strong immune response against tumors. This suggests that G-LNPs-based mRNA vaccines will be a powerful and flexible way to treat cancer [98].

LNPs have two benefits for mRNA vaccine vector: first, it protects mRNA from being broken down by endosomal enzymes [99], which improves delivery efficiency [100]. Second, it makes it easier for mRNA to be delivered for expression through a variety of biological processes, showing great biocompatibility. Therefore, experts view LNP as a promising mRNA delivery vector due to its excellent biocompatibility and high delivery efficiency. Currently, mRNA vaccines and therapeutics targeting COVID-19 and influenza extensively utilize LNP [101, 102].

Xu et al. developed the AGILE platform, integrating deep learning and combinatorial chemistry, to accelerate the development of ionizable lipids. Through virtual screening and experimental validation, two new lipids, H9 and R6, were discovered, which exhibited excellent performance in muscles and macrophages respectively, demonstrating the great potential of the combination of deep learning and combinatorial chemistry, and providing a new tool for personalized delivery of mRNA therapy. By optimizing the lipid structure and formulation, AGILE significantly improved the efficiency and specificity of mRNA delivery, opening up new avenues for vaccines and gene therapy [103]. In addition, Witten J et al. [104] also developed a deep learning (DL)-based method, LiON (Lipid Optimization using Neural Networks), to accelerate the design of novel ionizable lipids and enhance their delivery efficiency in different tissues. Through model prediction and experimental validation, two highly efficient lipid structures, FO-32 and FO-35, were discovered. They show great potential in muscle injection vaccines, nasal vaccines, and lung gene therapy. Particularly, the whole lung transfection ability of FO-32 and FO-35 is of great significance in the treatment of obstructive lung diseases such as cystic fibrosis. Vander Straeten A et al. successfully developed an automated microneedle vaccine printer (MVP) device, which can efficiently produce microneedle patches (MNP) containing LNP mRNA vaccines. The MNPs can be stably stored at room temperature for 6 months and still induce a strong immune response [105].

The exposure of the mRNA to the water at the core of an LNP makes the mRNA very vulnerable, and the instability of LNP mRNA has become a key factor affecting the quality of the formulation. Compared with the LNP mRNA system, the siRNA-LNP drug Onpattro™ has a significantly longer shelf life, suggesting that the current stability bottleneck is not the LNP but the mRNA. Lyophilization, as an effective preservation method, avoids the protein denaturation and loss of biological activity that may occur under high temperatures in traditional drying methods. Therefore, it is considered particularly suitable for the long-term preservation of biological macromolecular products, such as mRNA vaccines and enzyme preparations [106]. Munoz-Moreno R et al. compared the stability of freeze-dried formulations (stored at 5 °C) and freeze formulations (stored at -70 °C). The experimental data showed that in the 24-month stability test, the EC50 value (the in vitro expression of half-effective concentration) of the freeze-dried formulation was close to that of the freeze formulation (the freeze-dried version was only slightly higher), indicating that the freeze-drying process had a negligible impact on the structural integrity of mRNA. The immunogenicity test revealed that the level of gE-specific IgG antibodies induced by the freeze-dried vaccine was not significantly different from that of the freeze formulation, and was significantly higher than that of the control group [107].

In the phase I clinical study, the cytomegalovirus vaccine developed by Moderna (mRNA-1647) was in the form of a freeze-dried liquid formulation. However, in the subsequent phase II study, the formulation of this vaccine changed to a freeze-dried form. According to relevant reports, the freeze-dried formulation can be stably stored for up to 18 months under refrigeration conditions, and it can be clearly observed that the freeze-dried formulation has significant advantages in improving the stability of the formulation. Additionally, BioNTech/Pfizer is also conducting a phase III clinical trial to evaluate the freeze-dried version of its BNT162b2 vaccine. Recently, Reckitt Benckiser successfully developed the RH109 lyophilized mRNA vaccine [108], which has entered the clinical phase. The company developed three freeze-dried vaccines targeting the wild-type, Delta, and Omicron SARS-CoV-2 variants. These vaccines demonstrated efficacy in inducing high IgG titers and neutralization responses alongside long-term stability during storage at both 4 °C and 25 °C. This stability profile suggests potential to address the storage and transportation challenges associated with current mRNA vaccines. However, the specific stabilizing excipients used in the formulation and the detailed lyophilization protocols were not disclosed by the company.

Freeze-drying technology has become an important tool in modern vaccine research and global distribution by enhancing vaccine stability, simplifying storage conditions, ensuring efficacy and reducing costs. With the rapid development of nucleic acid vaccines such as mRNA, further optimization of the freeze-drying process (such as rapid freeze-drying and the application of new protective agents) will continue to drive the innovation of vaccine technology.

Nanostructured lipid carrier (NLC)

The use of NLC facilitated the successful development of the world's first saRNA vaccine, ARCT-154, for adult COVID-19. This vector can maintain the vaccine in a freeze-dried state, with a shelf life of up to 21 months under refrigeration and 8 months at ambient temperature. Furthermore, the utilization of this vaccine has markedly enhanced its thermal stability. It can keep being immunogenic for at least 6 months at room temperature or under normal refrigeration after being lyophilized, which solves a big technical problem in making RNA vaccines. This saRNA vaccine also works well to make the SARS-CoV-2 spike protein antigen, which causes a strong humoral and cell-specific immune response in mice without the need for nucleoside modification. For the ARCT-154 saRNA vaccine, the lowest dose needed is only 5 µg [72]. For comparison, the Moderna mRNA vaccine needs 100 µg and the BioNTech/Pfizer mRNA vaccine needs 30 µg. In conclusion, NLC delivery vehicles demonstrate the capability to enhance RNA technology and facilitate the broader application of RNA vaccines for the ongoing COVID-19 pandemic and future infectious disease pandemics [109].

Polymers

Due to their excellent physical stability, polymer carriers find extensive use in mRNA distribution. Two primary categories of polymer delivery vehicles are polyethylenimine (PEI) and poly-(β-amino ester) (PBAE). Currently, PEI serves as the primary delivery method for mRNA vaccines [110–112]. PEI exhibits early cytotoxicity attributable to its elevated molecular weight, thereby constraining its applicability. Research indicates that the amalgamation of PEI with PEG or cyclodextrin can diminish the positive surface charge and mitigate cytotoxicity [113]. A 2007 study showed that PBAE may effectively transfect mRNA in serum-free protein systems [114]. However, a study in 2007 revealed its potential for mRNA delivery, particularly in a serum protein-free system where PBAE achieved more efficient mRNA transfection in vitro [115]. This discovery has facilitated a broader range of studies and uses of PBAE. Researchers Kaczmarek et al. made a polymer-based delivery system

in 2016 called a PBAE and lipid–PEG complex. This system showed a lot of stability in serum and good mRNA delivery. Following intravenous administration, the mouse lungs effectively identified the target mRNA product [116].

Protamine

Drug delivery, diagnostics, and vaccine development use peptide- or protein-based biomaterials because of their biocompatibility and availability [117, 118]. Positively charged cationic peptides readily transport negatively charged mRNA through electrostatic interactions. Cationic peptides facilitate the delivery of negatively charged mRNA via electrostatic interactions, wherein the positive charge originates from lysine and arginine residues, enabling the mRNA to adhere firmly to the cationic peptide [119]. Protamine, a cationic peptide, establishes a stable compound with mRNA via electrostatic interactions, thereby safeguarding it from RNase-mediated destruction [120]. However, the tight binding between protamine [41] and mRNA may inhibit the translation process when using only protamine-formulated RNA [121]. However, CureVac's RNActive technology can circumvent this issue, the RNActive® vaccine technology effectively addresses the issue of protamine–mRNA complex inhibiting translation through a dual-component system. Its core technical principle includes: exposed optimized mRNA and protamine–mRNA complex. Specifically: the exposed mRNA achieves a 4–fivefold enhancement in expression through sequence optimization (UTR modification, codon optimization, etc.), while the protamine complex mainly fulfills the immune adjuvant function mediated by TLR7/8. The optimized mRNA is combined with fish protamine at a specific ratio to form a dual-component vaccine system. The naked mRNA enters the cytoplasm through scavenger receptors for efficient antigen expression; the protamine complex enters the endosome through endocytosis and activates innate immunity through TLR7/8. This innovative design not only overcomes the inhibitory effect of the simple protamine complex on translation but also realizes self-adjuvantic activity [122]. It has shown superior protective efficacy to traditional mRNA vaccines in preclinical models of influenza vaccines and tumor vaccines [123, 124].

Mechanisms of immune response induced by mRNA vaccines

Once mRNA vaccines enter cells, they serve as pathogen-associated molecular patterns (PAMPs) that are recognized by the pattern recognition receptors (PRRs) of the host cell [125]. PRRs bind to specific ligands, triggering type I interferon and NF-κB signaling pathways, which in turn mediate innate and adaptive immune responses.

The most notable PRRs that recognize exogenous RNA include TLRs located on endosomes and retinoic acid-inducible gene I-like receptors (RLRs) found in the cytoplasm. The TLR family consists of TLR3, TLR7, and TLR8. TLR3 is capable of recognizing and binding double-stranded RNA (dsRNA) that is more than 45 base pairs in length, as well as single-stranded RNA (ssRNA) with secondary structure or dsRNA derived from viral replication intermediates. This process regulates the activation of the IFN-I pathway and the secretion of cytokines and chemokines [126]. TLR7 can bind both dsRNA and ssRNA, and its activation process can enhance antigen presentation, promote cytokine secretion, and stimulate B cell responses [127]. In comparison to TLR3 and TLR7, TLR8 is limited to ssRNA [128]. RLRs consist of RIG-I and Melanoma differentiation-related gene 5 (MDA5) [129]. RIG-I primarily detects the 5'-terminal triphosphate (5'-ppp) of dsRNA and stimulates the production of IFN-I [130–132]. MDA5 is a cytosolic RNA sensor that can recognize long dsRNA produced during RNA virus replication and RNA transcription in vitro [133]. However, research has demonstrated that the dsRNA-activated protein kinase receptor (PKR) and oligoadenylate synthetase (OAS) pathway are activated, leading to the inhibition of mRNA translation [134] and the promotion of RNA degradation [135]. Therefore, it is crucial to purify mRNA.

Purified mRNA, which contains only ssRNA, is endogenously taken up by the delivery system (Fig. 2). It is then released to endosomes and translated to protein by ribosomes. The translated proteins activate the immune

system in two main ways: First, they are proteasome proteins that turn into peptides, which then combine with major histocompatibility complex (MHC) class I molecules on the cell surface antigen. These molecules bind with T cell receptors (TCRs), thereby activating CD8⁺T cells to kill infected cells by secreting perforin and granzyme. Second, exogenous antigens can be internalized by cells, processed into peptide fragments within cellular compartments, and presented via MHC class II molecules on the cell surface to CD4⁺T cells. CD4⁺T cells can either activate cellular immune response through secretion of inflammatory cytokines or activate humoral immune response through mutual activation of B cells [136].

Research and clinical trials on mRNA vaccines for infectious diseases

COVID-19

At the conclusion of 2019, the emergence of the novel coronavirus precipitated widespread infection and mortality, presenting a significant threat to the global community. In response to this problem, the mRNA-1273 vaccine developed by Moderna and the BNT162b2 mRNA vaccine created by BioNTech in partnership with Pfizer received emergency authorization from the US FDA in December 2020. Nonetheless, SARS-CoV-2 persists in mutating, and the infection incidence remains elevated. Research aimed at the development of efficacious pharmaceuticals and vaccines continues to advance swiftly. Currently, multiple mRNA vaccines have commenced clinical trials (Table 1).

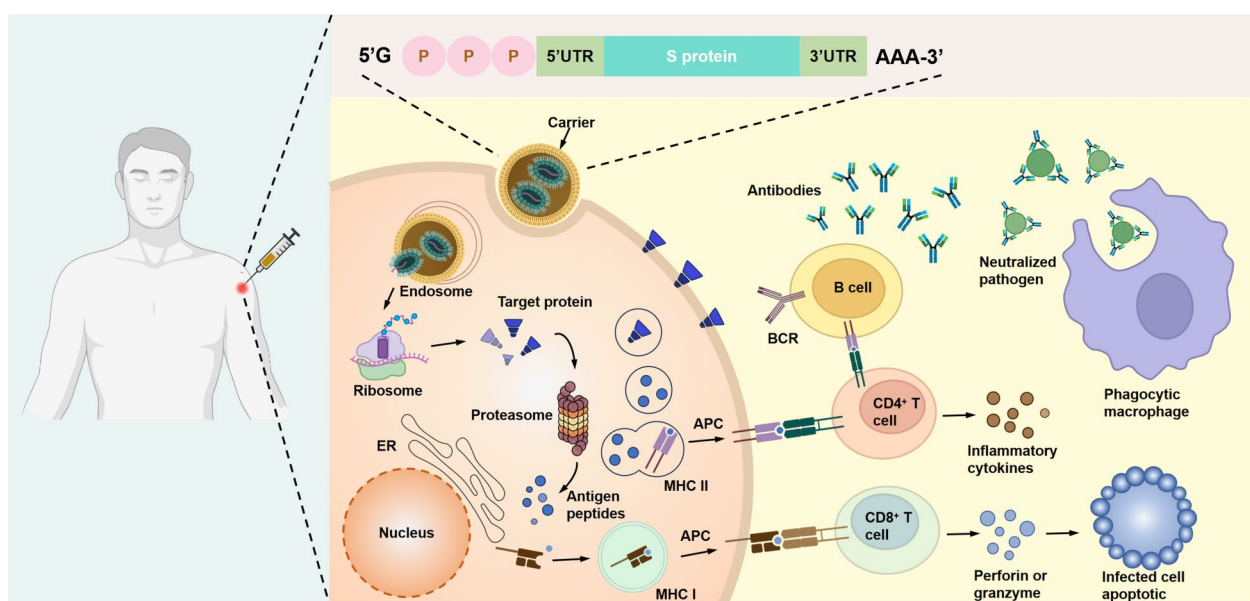


Fig. 2 Mechanism of cellular and humoral immune responses induced by mRNA vaccines

Table 1 Clinical application of COVID-19 mRNA vaccine

Type of candidate vaccine	Types of RNA	Trial characteristics	Developers	Phase	Number
mRNA-1273 (Spikevax)	mRNA	2 doses; Day 0 + 28; IM	Moderna + National Institute of Allergy and Infectious Diseases (NIAID)	Phase 4	NCT04760132
BNT162b2 (3 LNP-mRNAs), also known as "Comirnaty"	mRNA	2 doses; Day 0 + 21; IM	Pfizer/BioNTech + Fosun Pharma	Phase 4	NCT04674189
CVnCoV Vaccine	mRNA	2 doses; Day 0 + 28; IM	CureVac AG	Phase 3	NCT04760132
ARCT-021	saRNA	NR; NR; IM	Arcturus Therapeutics	Phase 2	NCT04668339
LNP-nCoVsaRNA	saRNA	2 doses; NR; IM	Imperial College London	Phase 1	ISRCTN17072692
SARS-CoV-2 mRNA vaccine (ARCoV)	mRNA	2 doses; Day 0 + 14 or Day 0 + 28; IM	Academy of Military Science (AMS), Walvax Biotechnology and Suzhou Abogen Biosciences	Phase 3	NCT04847102
ChulaCov19 mRNA vaccine	mRNA	2 doses; Day 0 + 21; IM	Chulalongkorn University	Phase 2	NCT05605470
PTX-COVID19-B, mRNA vaccine	mRNA	2 doses; Day 0 + 28; IM	Providence Therapeutics	Phase 3	NCT05534035
CoV2 SAM (LNP) vaccine. A self-amplifying mRNA (SAM) lipid nanoparticle (LNP) platform + Spike antigen	saRNA	2 doses; Day 0 + 30; IM	GlaxoSmithKline	Phase 1	NCT04758962
mRNA-1273.351	mRNA	3 doses; Day 0 or Day 0 + 28 or Day 56; IM	Moderna + National Institute of Allergy and Infectious Diseases (NIAID)	Phase 4	EUCTR2021-000930-32
MRT5500, an mRNA vaccine candidate	mRNA	2 doses; Day 0 + 21; IM	Sanofi Pasteur and Translate Bio	Phase 2	NCT04798027
DS-5670a, coronavirus-modified uridine RNA vaccine (SARS-CoV-2)	mRNA	2 doses; NR; IM	Daiichi Sankyo Co., Ltd	Phase 2/3	JPRN-jRCT2071210106
HDT-301: Self-replicating mRNA vaccine formulated as a lipid nanoparticle	saRNA	2 doses; Day 0 + 28; IM	SENAI CIMATEC	Phase 2/3	NCT05542693
mRNA-1283	mRNA	2 doses; Day 0 + 28; IM	ModernaTX, Inc	Phase 1	NCT04813796
EXG-5003; a temperature-sensitive self-replicating RNA vaccine expressing the receptor binding domain of the SARS-CoV-2 spike protein	saRNA	1 dose; Day 0; ID	Elixirgen Therapeutics, Inc	Phase 1/2	NCT04863131
mRNA COVID-19 vaccine (SW-BIC-213)	mRNA	2 doses; TBD; IM	Shanghai East Hospital and Stemirna Therapeutics	Phase 3	NCT05580159
LNP-nCoV saRNA-02 vaccine; Self-amplifying RNA (saRNA) encapsulated in lipid nanoparticles (LNP)	saRNA	2 doses; Day 0 + 28; IM	MRC/UVRI and LSHTM Uganda Research Unit	Phase 1	NCT04934111
mRNA-1273.211. A multivalent booster candidate combining mRNA-1273 plus mRNA-1273.351	mRNA	1 dose; Day 0; IM	ModernaTX, Inc	Phase 2/3	NCT04927065
ARCT-154 mRNA Vaccine	saRNA	2 doses; Day 0 + 28; IM	Arcturus Therapeutics, Inc	Phase 3	ISRCTN15779782
ARCT-165 mRNA Vaccine	saRNA	2 doses; Day 0 + 29; IM	Arcturus Therapeutics, Inc	Phase 1/2	NCT05037097
ARCT-021 mRNA Vaccine	saRNA	2 doses; Day 0 + 30; IM	Arcturus Therapeutics, Inc	Phase 1/2	NCT05037097
HDT-301 vaccine	saRNA	1–2 doses; Day 0 ± 56; IM	HDT Bio	Phase 1	NCT05132907
VLP-01, self-amplifying RNA vaccine against the coronavirus	saRNA	2 doses; NR; IM	VLP Therapeutics Japan GK	Phase 1	jRCT2071210067
EG-COVID vaccine	mRNA	3 doses; Day 0 + 21 + 42; IM	EyeGene Inc	Phase 1/2	NCT05188469
Coronavirus mRNA vaccine (LVRNA009)	mRNA	2 doses; Day 0 + 28; IM	AIM Vaccine and Liverna Therapeutics	Phase 3	NCT05428592
mRNA-1273.529—Booster	mRNA	1 dose; Day 0; IM	ModernaTX, Inc	Phase 2/3	NCT05249829
CV2CoV, mRNA vaccine	mRNA	1 dose; Day 0; IM	CureVac AG	Phase 1	NCT05260437

Table 1 (continued)

Type of candidate vaccine	Types of RNA	Trial characteristics	Developers	Phase	Number
mRNA vaccine (MIP-SCo-mRNA-RBD-1)	mRNA	1 dose; Day 0; IM	University of Melbourne	Phase 1	NCT05272605
A Lyophilized COVID-19 mRNA Vaccine	mRNA	1 dose; Day 0; IM	Jiangsu Rec-Biotechnology Co., Ltd	Phase 1	NCT05351450
COVID-19 mRNA Vaccine (SYS6006)	mRNA	2 doses; Day 0 + 21; IM	CSPC ZhongQi Pharmaceutical Technology Co, Ltd	Phase 2	NCT05439824
mRNA GEMCOVAC-19 (COVID-19 vaccine)	mRNA	2 doses; Day 0 + 28; IM	Gennova Biopharmaceuticals Limited	Phase 2/3	CTRI/2022/04/041880
Lyophilized COVID-19 mRNA Vaccine	mRNA	1 dose; Day 0; IM	Wuhan Recogen Biotechnology Co., Ltd	Phase 1	NCT05366296
A self-amplifying RNA (saRNA) boost vaccines (AAHI-SC2 and AAHI-SC3)	saRNA	1 dose; Day 0; IM	ImmunityBio, Inc	Phase 1/2	NCT05370040
RQ3013: SARS-CoV-2 mRNA Chimera Vaccine	mRNA	1 dose; Day 0; IM	Walvax Biotechnology; Shanghai RNACure Biopharma	Phase 1	NCT05396573
mRNA-1273.214 (Booster)	mRNA	2 doses; Day 0 + 55; IM	ModernaTX	Phase 3	NCT05436834
mRNA-1073; (COVID-19/Influenza) Vaccine	mRNA	2 dose; Day 0; IM	ModernaTX	Phase 1/2	NCT05375838
RVM-V001	mRNA	1 dose; Day 0; IM	RVAC Medicines	Phase 1	NCT05420077
ABO1009-DP (COVID-19 Omicron) mRNA Vaccine; ABO1020 vaccine	mRNA	1 dose; Day 0; IM	Suzhou Abogen Biosciences Co., Ltd	Phase 1	NCT05433194
Self-Amplifying Messenger Ribonucleic Acid (samRNA) Vaccines	saRNA	2 doses; Day 0 + 28; IM	Gritstone bio, Inc	Phase 1	NCT05435027
Investigational CV0501 mRNA COVID-19 Vaccine	mRNA	1 dose; Day 0; IM	GlaxoSmithKline	Phase 1	NCT05477186
GLB-COV2-043, an mRNA booster vaccine candidate	mRNA	1 dose; Day 0; IM	GreenLight Biosciences, Inc	Phase 1/2	NCT05602961
mRNA-based COVID-19 vaccine (CReNAPCIN)	mRNA	1 dose; Day 0; IM	ReNAP Technology	Phase 1	IRCT20230131057293N1
JCXH-221, an mRNA-based	mRNA	1 dose; Day 0; IM	Immorna Biotherapeutics, Inc	Phase 1/2	NCT05743335

*The data are sourced from <https://www.who.int/publications/m/item/draft-landscape-of-covid-19-candidate-vaccines> (25 December 2024)

Moderna's mRNA-1273 encodes the spike protein of SARS-CoV-2, encapsulated in LNP. It commenced phase I clinical trials in under 10 weeks following the publication of the initial SARS-CoV-2 genome sequencing, an unparalleled occurrence in the pharmaceutical business [137]. A phase I clinical investigation (ClinicalTrials.gov: NCT04283461) involved 45 healthy adults aged 18 to 55 years who received varying doses of the vaccine (50, 100, and 250 µg). Regardless of age, this study revealed a dose-dependent antibody response among subjects following two vaccine doses. Over fifty percent of the subjects encountered systemic adverse reactions, particularly in the high-dose group (250 µg), with the prevalence of severe adverse events reaching 21%. Despite the decline in neutralizing and binding antibody levels over time, they persisted at a comparatively elevated level [138]. A phase II clinical trial (ClinicalTrials.gov: NCT04405076) showed that after the second dose of the vaccine, the 100 µg dose caused higher levels of binding

and neutralizing antibodies. During the study, no major side effects were reported. The results validate the safety and immunogenicity of 50 µg and 100 µg mRNA-1273 in a two-dose immunization protocol [139]. Over 30,000 volunteers participated in the phase III clinical investigation (ClinicalTrials.gov: NCT04470427). Among the participants, 2.2% had prior SARS-CoV-2 infections before the experiment commenced. The findings indicated that significant adverse effects were infrequent following two doses of the vaccination. Furthermore, the vaccination demonstrated 94.1% efficacy in preventing COVID-19 infection [140]. In February 2021, an extensive phase IV clinical trial (ClinicalTrials.gov: NCT04760132) was initiated to more accurately assess the vaccine's efficacy and safety.

The BNT162b2 vaccine is made up of an LNP-formulated nucleoside-modified mRNA that codes for the full trans-membrane S protein in its pre-fusion conformation (S-pp). A phase I/II study (ClinicalTrials.gov:

NCT04380701) with healthy people ages from 19 to 55 showed that the BNT162b2 vaccine worked to produce both antibody and T cell responses after the first dose, with booster doses ranging from 1 to 30 µg. When compared to the titer in plasma from people who had a booster shot seven days later, the geometric mean titer (GMT) of SARS-CoV-2 serum 50% neutralization ranged from 0.3-fold (1 µg dose group) to 3.3-fold (30 µg dose group). The study's results indicated that immunization with BNT162b2 at a tolerable dosage can elicit an adaptive mixed humoral and cellular immune responses, which collectively leads to protection against COVID-19 [141]. The outcomes of the phase III clinical trial (ClinicalTrials.gov: NCT04368728), which included over 30,000 participants, demonstrated the significant efficacy of the BNT162b2 vaccine. The vaccine demonstrated 52% efficacy following the initial dose and 95% efficacy after the administration of two doses in preventing mild to moderate SARS-CoV-2 infection. This research indicated that the occurrence of severe adverse events post-vaccination was minimal and not substantially different from the placebo cohort, demonstrating the vaccine's safety [142]. The phase IV study (ClinicalTrials.gov: NCT04844489) commenced in February 2021. This is a longitudinal study to assess the efficacy and longevity of the BNT162b2 mRNA vaccine and the safety of SARS-CoV-2 inoculation in the general populace.

The global advancement of COVID-19 mRNA vaccines is vigorously encouraged, with numerous vaccine candidates undergoing different phases of clinical trials. CanSino Biologics Inc. is undertaking phase I and II clinical trials (ClinicalTrials.gov: NCT05373485 and NCT05373472) of a COVID-19 mRNA vaccine (LNP) to assess its safety and immunogenicity in adults aged 18 and older. A total of 300 participants participated in these two studies, categorized into low-dose (30 µg) and high-dose (50 µg) groups, and vaccinated on days 0 and 21, respectively. Nevertheless, the outcomes of these experiments remain unpublished. CureVac has started a phase II/III trial (NCT04652102) to test the CVnCoV SARS-CoV-2 mRNA vaccine (LNP) candidate to see how well it works and how safe it is. The trial enrolled 39,693 individuals, with the experimental group administered 12 µg of the CVnCoV vaccination intramuscularly on days 1 and 29. The findings indicated that the vaccination demonstrated efficacy of 70.7% and 77.2% against mild-to-severe novel coronavirus pneumonia in individuals aged 18 to 60. This affirmed that the two-dose CVnCoV vaccination regimen is efficacious in preventing symptomatic COVID-19 in adults and possesses a dependable safety profile [143]. SinoMicro has commenced phase I/II clinical research to assess the safety, immunogenicity, and immunological durability of the COVID-19 mRNA

vaccine (LNP) in healthy individuals aged 18 and above. The trial enlisted 480 participants, categorized into a high-dose group (45 µg) and a low-dose group (25 µg), who got intramuscular injections every 29 days. The clinical trial has concluded, but its results remain unpublished. The outcomes of these clinical trials are essential for assessing the efficacy and safety of mRNA vaccines in preventing COVID-19 and will furnish significant scientific data to bolster the global response to the pandemic.

Influenza

Among the most extensively researched vaccinations are mRNA vaccines targeting influenza viruses. Owing to the elevated mutation rate of influenza viruses, vaccine research and development must proceed in tandem with the evolving viral strains; hence, mRNA vaccines are particularly advantageous. For instance, during the lethal H7N9 influenza virus strain epidemic in China in 2013, researchers successfully cloned the HA gene into the pDNA template of the self-amplifying mRNA vaccine, leading to the successful production of the saRNA vaccine within 8 days of the HA gene sequence's release [144] (Supplementary Table 1).

In 2017, Moderna executed a pioneering trial to assess the safety and immunogenicity of mRNA vaccines VAL-506440 and VAL-339851 against the potentially pandemic avian influenza viruses H10N8 and H7N9 for the first time (ClinicalTrials.gov: NCT03076385, NCT03345043). VAL-506440 and VAL-339851 mRNA vaccines consisted of chemically modified mRNAs encoding the full-length, membrane-bound form of the hemagglutinin (HA) glycoprotein from the H10N8 influenza strain (A/Jiangxi-Donghu/346/2013) or the H7N9 influenza strain (A/Anhui/1/2013) [155], an LNP delivery system was used [104]. A single facility in Germany conducted two randomized, placebo-controlled, double-blind Phase I clinical trials for H10N8 and in the United States for H7N9. The H10N8 vaccine research comprised 201 participants aged from 18 to 64 years, whereas the H7N9 vaccine study involved 156 participants aged from 18 to 49 years. Participants were administered two doses of either the vaccination or a placebo, spaced three weeks apart. The H10N8 vaccine trial assessed intramuscular doses ranging from 25 to 400 µg and intradermal doses of 25 and 50 µg. All the people who were given the 100 µg dose showed hemagglutination inhibition (HAI) levels of 1:40 or higher, and 87.0% of them showed microneutralization (MN) levels of 1:20 or higher. Zero. The 25 µg intradermal dosage elicited HAI titers exceeding 1:40 in 64.7% of subjects. The H7N9 vaccine trial assessed intramuscular dose levels of 10, 25, and 50 µg. This study showed that injecting 10, 25, or 50 µg of the drug into the muscle caused HAI titers of 1:40 or more in 36.0%,

96.3%, and 89.7% of the subjects, respectively. The 10 µg and 25 µg doses attained 100% MN titers in all subjects, whereas the 50 µg dose group reached 96.6%. Both H10N8 and H7N9 mRNA intramuscular vaccinations had favorable safety profiles and immunogenicity, with no major side events linked to the vaccines described. These experiments represent the inaugural mRNA vaccines targeting the H10N8 and H7N9 influenza viruses. They are not only well tolerated but also elicit a robust humoral immune response, offering a novel method for the prevention of future pandemic influenza [145].

Rabies

Rabies is a critical, acute zoonotic infectious illness responsible for around 59,000 fatalities annually. The rabies virus (RABV), consisting of structural proteins and negative-stranded RNA, is the causal agent of the disease. Glycoproteins are the principal component of RABV and serve as its primary antigen, eliciting a particular immunological response in the host. Rabies is highly virulent and exhibits a dismal prognosis once it manifests. At present, vaccination is the most efficacious method for preventing and managing the dissemination of the rabies virus (Supplementary Table 1).

Recently, CureVac initiated a proof-of-concept human clinical trial to investigate an mRNA-based rabies vaccine. CV7201 is an unmodified, yet sequence-optimized mRNA molecule encoding a rabies virus glycoprotein in which the core delivery system is the protamine complex (ClinicalTrials.gov: NCT02241135). This research was done for the first time in healthy adults, recruiting participants aged 18 to 40 years, all of whom had not received prior rabies vaccinations. The participants were delivered three doses of the CV7201 vaccine, with dosages varying from 80 to 640 µg, via either intradermal or intramuscular injection using standard syringes with needles and three different types of needle-free devices. In the latter group of vaccinated individuals, certain volunteers received a booster dosage one year afterward. The majority of participants reported mild to moderate injection-site reactions post-vaccination, while 78% exhibited systemic adverse effects, including fever, exhaustion, and pain. The trial results showed that the needle-injected vaccine failed to elicit a sufficient antibody response at various dosages and administration modes. Still, the mRNA vaccine candidate worked well to make a strong immune response against viral antigens when given through a needle-free device. The vaccination showed satisfactory safety and received adequate tolerance [146]. CureVac initiated a phase I clinical investigation (ClinicalTrials.gov: NCT03713086) to assess the safety, reactogenicity, and immunogenicity of a low-dose unmodified mRNA rabies vaccine, CV7202. This research involved

55 healthy adults aged from 18 to 40 years who received intramuscular injections of 1 µg, 2 µg, or 5 µg of CV7202 on days 1 and 29, respectively. The results showed that both the 1 µg and 2 µg doses of the CV7202 vaccine were well tolerated and caused a rabies-neutralizing antibody response in all of the people who were vaccinated, which was in line with WHO guidelines. Nonetheless, the 5 µg dose of the vaccine exhibited suboptimal reactogenicity [147].

Zika virus

The Zika virus (ZIKV), identified in 1947, is a member of the flavivirus family and transmitted via mosquitoes [148]. Recent research indicates that ZIKV is linked to congenital anomalies and neurological conditions, including microcephaly in infants and Guillain-Barré syndrome in adults [149]. Researchers led by Norbert Pardi and others have shown that nucleoside-modified mRNA-LNP vaccines targeting the pre-membrane and envelope (prM-E) glycoprotein of the 2013 ZIKV outbreak strain can produce a strong and long-lasting neutralizing antibody response after a single low-dose intradermal immunization in both mice and nonhuman primates. This shows that nucleoside-modified mRNA-LNP vaccines can quickly build up long-lasting immunity and are a good way to control ZIKV around the world [150]. The modified mRNA vaccine that Justin M. Richner et al. created effectively protects against ZIKV-related illness and can be customized to make a person less likely to get dengue virus (DENV) in future [94]. Phase I clinical trials conducted by Moderna to assess the safety and immunogenicity of mRNA-1325 (ClinicalTrials.gov: NCT03014089) and mRNA-1893 (ClinicalTrials.gov: NCT04064905) against Zika virus indicated that mRNA-1893 showed superior efficacy compared to mRNA-1325 [151]. Previous animal model studies demonstrated that the neutralizing antibody titer of mRNA-1893 was approximately 20 times greater than that of mRNA-1325 [152]. A phase II trial concerning mRNA-1893 is now in progress.

Human immunodeficiency virus (HIV)

HIV is a persistent virus that can integrate into the host cell's DNA, remaining for the lifespan of the infected cell. HIV infection poses a significant threat to world health, with almost 2 million new adult infections annually. Consequently, the advancement of secure and efficacious HIV preventive strategies is essential to mitigating the global AIDS epidemic. Researchers have already shown that adding nucleosides to the HIV-1 envelope protein (Env) 1086C and changing it could make the mRNA-LNP vaccine work better against cells in the serum of rabbits and rhesus monkeys after they were vaccinated

[153]. Moreover, mRNA vaccines that co-express membrane-anchored HIV-1 Env and simian immunodeficiency virus (SIV) Gag proteins to generate virus-like particles (VLPs) can elicit the generation of broadly neutralizing antibodies and diminish the risk of infection in rhesus monkeys [154]. A lot of research has shown that NanoVac formulations work better than LNP vectors at lowering the amount of HIV-1 virus in the body [155]. New research shows that using germline-targeted priming-boost immunogens along with mRNA-LNP technology can create progenitors of broadly neutralizing antibodies BG18 against the V3-glycan region of HIV-1. This is an innovative way to make HIV vaccines more effective [156]. HIV mRNA vaccines have initiated numerous clinical trials (Supplementary Table 1). The results of the first phase of iHIVARNA-01's clinical study, which was started by Judit Pich Martínez (ClinicalTrials.gov: NCT02413645), showed that iHIVARNA was safe, well tolerated, and able to cause a moderate HIV-specific T cell response [157]. The phase II clinical trial (ClinicalTrials.gov: NCT02888756) proved that iHIVARNA-01, a possible vaccine that could be used along with or instead of cART for people with chronic HIV infection, was safe [158]. The US National Institute of Allergy and Infectious Diseases finished a phase I clinical trial (ClinicalTrials.gov: NCT05217641) with healthy, uninfected adults to test the safety and immunogenicity of an HIV trimer mRNA vaccine. The results have not been made public yet. The International AIDS Vaccine Initiative paid for a separate phase I trial that looked at the safety and immunogenicity of the eOD-GT8 60-polymer mRNA vaccine (mRNA-1644) and the Core-g28v2 60-polymer mRNA vaccine (mRNA-1644v2-Core) (ClinicalTrials.gov: NCT05217641). The results have not been made public yet.

Tuberculosis

TB is an infectious disease caused by *M.tb*. Currently, the BCG vaccine remains the only approved anti-TB vaccine and has shown significant protective effects on infants and young children [159]. However, its effectiveness in protecting against pulmonary TB in adults is limited [160–162].

In 2004, the Tascon RE team at the National Institute for Medical Research in London, United Kingdom, conducted the first experimental study of an mRNA vaccine against TB. This study involved using nude IVT mRNA encoding the immunodominant antigen MPT83 to immunize mice (four times with an interval of 3 weeks). The researchers successfully induced a protective immune response against *M.tb* challenge in a short period of time [163]. In 2010, the Coelho-Castelo AA team of the University of Sao Paulo, Brazil, extracted the mRNA

of Hsp65 protein from *Mycobacterium leprae*. They inoculated mice with a single dose of 10 µg of mRNA-Hsp65 through the nasal route. Two weeks after immunization, the mice were euthanized. The final experimental results demonstrated that this method could induce protective immune responses in the mouse TB model and reduce the bacterial load and inflammatory response in the lungs of mice [164]. In 2022, the Coler RN team at the Seattle Children's Research Institute in the United States, along with Seattle HDT Bio biotechnology companies, reported on the use of the replication rep RNA platform encoding ID91 TB vaccine candidate antigens in preprints on the BioRxiv platform. They utilized a first-generation unstructured NLC lipid carrier to coat ID91 and develop a rep RNA TB vaccine. In a mouse model, this vaccine demonstrated medium immunogenicity and protective effects. These findings represent the first evidence that the repRNA platform is a viable system for TB vaccination [165]. In addition, immunoinformatics methods can also be utilized in the development of TB mRNA vaccines. This includes epitope screening, construction of novel vaccines, and immune simulation, all of which will contribute to the further advancement of TB mRNA vaccines [166–168]. Currently, BioNTech is actively developing an mRNA vaccine for TB, and its research and development pipeline includes several vaccine candidates. Associated research has advanced to the phase I clinical trial stage (ClinicalTrials.gov: NCT05537038 and NCT05547464). The primary objective of these trials is to assess the safety and immunogenicity of the vaccine, with no particular data on protective effectiveness disclosed to date.

Conclusion, challenges, and future prospects

The successful development of a COVID-19 vaccine has showcased the significant potential of mRNA vaccine technology for disease prevention and control, surpassing traditional vaccinations. This achievement motivates global scientists to diligently create new mRNA vaccines for both infectious and non-infectious disorders, signaling a significant transformation in disease prevention and management. We anticipate a substantial rise in mRNA vaccine candidates and further advancements in clinical trials in the forthcoming years. The distinctive benefit of mRNA vaccine technology is in its capacity to encode several antigens, facilitating the advancement of a novel class of infectious disease vaccines. This enables the prevention of various infectious illnesses with a single vaccine, significantly enhancing the practicality and efficacy of vaccination.

Notwithstanding the numerous benefits of mRNA vaccines, they encounter some obstacles throughout the research and development phase. First, the production

process of mRNA vaccines faces three major technical challenges, which are the disadvantages of its application, such as the complex multi-step in vitro synthesis system, the dual threshold of equipment and technology, and the cost challenge of scale-up production. The production process involves enzymatic reactions, such as in vitro transcription of linearized plasmid DNA, 5' capping, and polyadenylation tail modification, as well as purification steps to remove impurities [49]. This process must establish a strict aseptic reaction system and precise molecular biology control standards. In addition, specific equipment, such as bioreactors, chromatography systems, and nano-liposome encapsulation devices, is required during the production process, and each production batch needs to undergo integrity testing, which increases the technical complexity. Moreover, from laboratory scale to commercial production, the reaction efficiency may decline, and the cumulative yield loss in the purification steps can reach 40%, resulting in a significantly higher unit dose cost than recombinant protein vaccines [169].

Second, mRNA vaccines synthesize in the cytoplasm and do not interact with human DNA, but their potentially robust immunogenic response may elicit a significant type I interferon and pro-inflammatory cytokine response, heightening the risk of inflammation and autoimmune disorders, particularly in the elderly, children, pregnant women, and individuals with chronic conditions such as autoimmune diseases. Therefore, preclinical and clinical trials must thoroughly evaluate the safety of vaccinations across various groups. Furthermore, various factors (such as pH value, temperature, and ionic concentration) can affect the stability of mRNA, making it difficult to maintain a stable state. Additionally, multiple factors—including pH, temperature, and ionic concentration—contribute to the limited stability of mRNA. It is worth noting that current research is breaking through process bottlenecks through the following directions: developing freeze-dried formulations, optimizing cap analog structures, and developing continuous flow production technologies, which are expected to improve the accessibility of mRNA vaccines in future.

Regarding the safety of mRNA vaccines and the known adverse immune reactions, Gurudeeban Selvaraj et al. reported [170] that the main reason for allergic reactions caused by mRNA vaccines lies in the presence of allergenic components in the vaccines, such as glycoproteins (the amino acid residue sequence 437–508 of polyphenolase) or stabilizers (such as sucrose and polyethylene glycol). These molecules can promote or stimulate the reaction between allergen-specific antibodies and IgE in the body, thereby triggering allergic inflammatory responses [171]. Allergens may also further lead to the release of inflammatory mediators

(including histamine, prostaglandins, leukotrienes, and proteases) and pro-inflammatory cytokines, thereby causing allergic reactions, which include nausea, vomiting, red skin, rash, laryngeal edema, wheezing, tachycardia, hypotension, and cardiovascular failure. Although significant reductions in the immunostimulatory effect of mRNA and enhancement of the immunogenicity of LNP were achieved through synthetic modification [172], when mRNA at a dose exceeding a certain level or certain ionizable lipids were administered in the form of LNP, it might trigger immune activation and cytotoxicity, thereby causing allergic reactions in individuals vaccinated with the vaccine. These reactions are mainly related to the lipid content and payload of LNPs. To address the immune stimulation caused by LNP mRNA, major clinical studies have focused on the use of the glucocorticoid receptor (GR) agonist dexamethasone [173] as well as corticosteroids to reduce the immune stimulation of mRNA [174]. Furthermore, an increasing number of clinical trials have confirmed the effectiveness and safety of mRNA vaccines in treating infectious diseases, providing valuable references for the development of vaccines for various diseases.

In conclusion, we should take advantage of the COVID-19 pandemic's impetus to conduct in-depth research on the properties and potential degradation mechanisms of mRNA vaccines in order to develop safe, stable, and effective mRNA vaccines against other bacterial infectious diseases (such as tuberculosis) and make groundbreaking advances in human and public health.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40001-025-03060-x>.

Additional file 1

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Author contributions

Y.L. and X.W. jointly conceptualized this scoping review. Z.Z. and J.D. and D.Z. extracted the data from the articles. Z.Z. and D.Z. cross-checked the extracted data. Z.Z. prepared the initial manuscript, Y.L., X.W. and R.H. edited the initial manuscript. All authors have read and agreed to the published version of the manuscript.

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